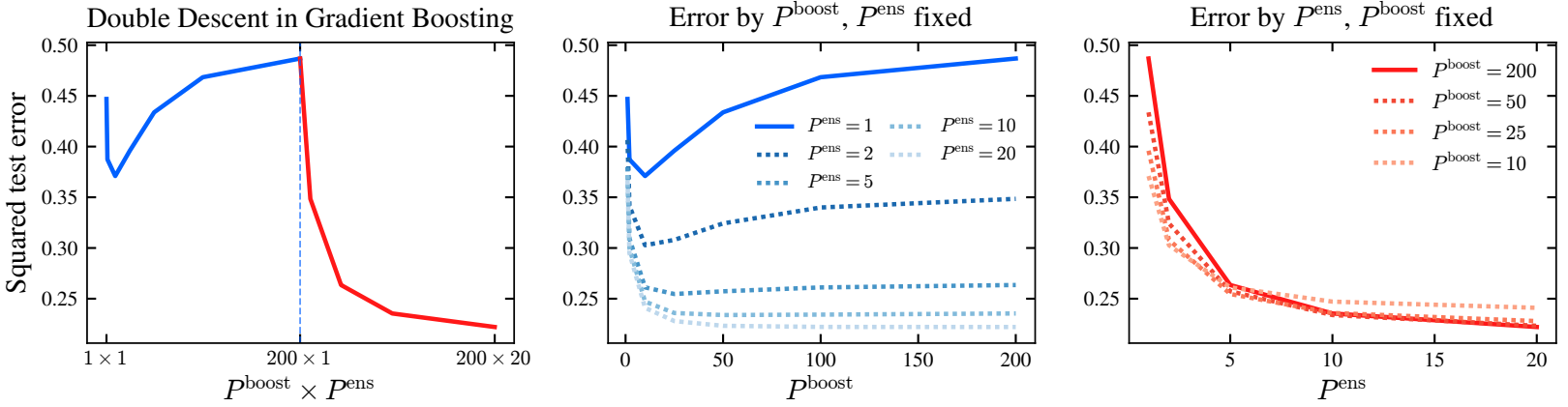
Double Descent in Gradient Boosting
Squared test error
0.50
0.45
0.40
0.35
0.30
0.25
1 × 1
200 × 1
200 × 20
$P^{\text{boost}} \times P^{\text{ens}}$
Error by P^{boost}, P^{ens} fixed
$P^{\text{ens}} = 1$
$P^{\text{ens}} = 2$
$P^{\text{ens}} = 5$
$P^{\text{ens}} = 10$
$P^{\text{ens}} = 20$
0
50
100
150
200
P^{boost}
Error by P^{ens}, P^{boost} fixed
$P^{\text{boost}} = 200$
$P^{\text{boost}} = 50$
$P^{\text{boost}} = 25$
$P^{\text{boost}} = 10$
5
10
15
20
P^{ens}